

Jul 28, 2023

Introduction to Computer and Network Security

Part 1

Question 1 [40 points]

In the context of cryptography, (a) state and describe the Kerckhoffs principle, (b) explain the purpose of key management, (c) explain the notion of symmetric key cryptography, and (d) explain the notion of asymmetric (or public) key cryptography.

Question 2 [10 points]

Describe the technique of k-anonymity and explain how it can mitigate linkage attacks.

Part 2

Question 3 [40 points]

In the context of the Diffie-Hellman key exchange, describe (a) the protocol, (b) the Man-In-the-Middle (MITM) attack, and (c) how it is possible to mitigate the MITM attack.

Question 4 [10 points]

Define the notions of vulnerability and threat and give (at least) an example for each one.

Please, remember...

Be sure to annotate the following information on each used sheet of paper (possibly, 2 sheets at most)

- Name
- Surname
- Student ID number
- Question number